Results: Hypotheses Explored

Results: Hypotheses Explored I

The template scores a configurable set of hypotheses about the topic. For this instance 6 hypotheses are declared in configuration; Table 7 lists them with their scope and evidence score.

Table 7. Hypotheses explored.

Results: Hypotheses Explored I

ID	Hypothesis	Scope	Evidence score
H1	Wakefulness Efficacy	clinical	+0.56
H2	Cognitive Enhancement	cognitive	+0.49
H3	Low Abuse Liability	safety	+0.62
H4	Dopaminergic Mechanism	pharmacological	+1.00
H5	Off-label Psychiatric Utility	applied	+0.35
H6	Tolerability	safety	+0.31

Results: Hypotheses Explored I

Evidence scores are produced by the optional, LLM-gated knowledge-graph stage. When the knowledge-graph stage is skipped (no language model configured), scores read *pending*. When the stage runs (as in this instance, with 127 assertions extracted via Ollama), scores are populated from citation-weighted assertion extraction. The hypotheses, their names, and their scope are always reported directly from configuration regardless of whether the LLM stage executed.

Interpretation I

Reported scores, when present, should be read as relative rankings rather than calibrated probabilities: absolute magnitudes are inflated by publication bias and the linguistic asymmetry of academic writing. A positive score indicates that the retrieved corpus *talks about* the hypothesis in a supporting direction; a negative score indicates contradicting evidence; a score near zero indicates either balanced evidence or insufficient coverage.

Interpretation I

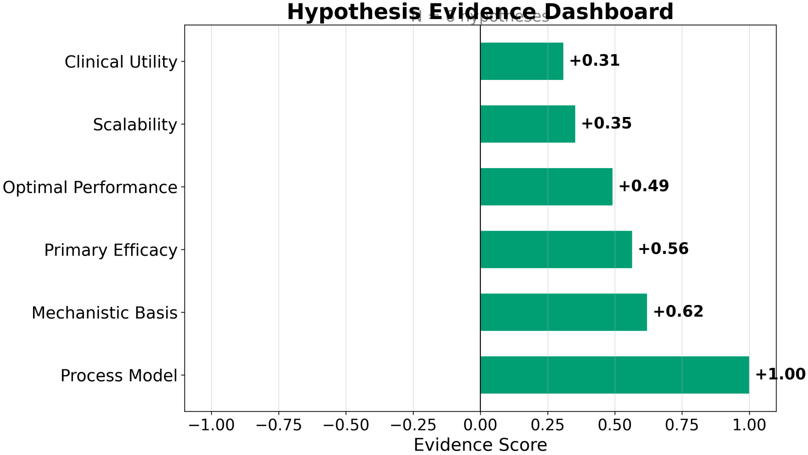
The six hypotheses frame the evidence landscape for Modafinil:

Interpretation I

- **H1 (Wakefulness Efficacy)** — the clinical claim that modafinil reliably promotes wakefulness in sleep-disorder populations. This is the primary indication and the most-studied claim in the corpus.
- **H2 (Cognitive Enhancement)** — the claim that modafinil improves attention, working memory, and executive function, especially under sleep deprivation. This hypothesis drives the neuroenhancement literature and is the subject of significant public and scientific debate.
- **H3 (Low Abuse Liability)** — the safety claim that modafinil has lower abuse potential than classical psychostimulants. This is critical for regulatory classification and prescribing decisions.
- **H4 (Dopaminergic Mechanism)** — the pharmacological claim that modafinil acts substantially via dopamine-transporter inhibition rather than a purely novel mechanism. This hypothesis has mechanistic and translational implications.
- **H5 (Off-label Psychiatric Utility)** — the applied claim that modafinil is a useful adjunct for fatigue and cognition in psychiatric and neurological conditions, including depression, ADHD, and schizophrenia.

Interpretation II

- **H6 (Tolerability)** — the safety claim that modafinil is generally well tolerated, with predominantly mild, transient adverse effects. This underpins its clinical acceptability relative to alternative wakefulness agents.



Interpretation III

Figure 1: Hypothesis dashboard for Modafinil. The dashboard summarizes the evidence scores across the 6 configured hypotheses, showing the direction and magnitude of citation-weighted assertion evidence.